

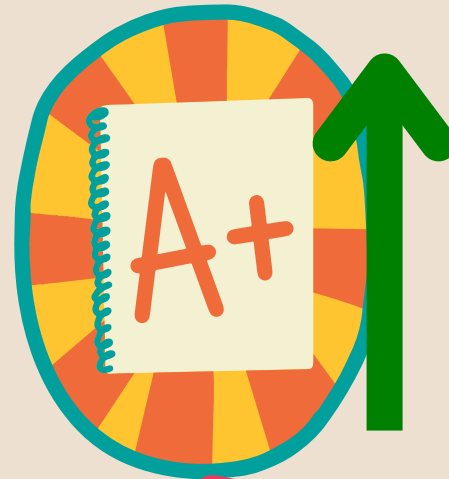
Clustering Students, Predicting Final Grades, and Optimizing LMS Feature Quantity Based on LMS Engagement Data

By Axel Arevalo

BACKGROUND



LEARNING
MANAGEMENT
SYSTEM (LMS)



CTL



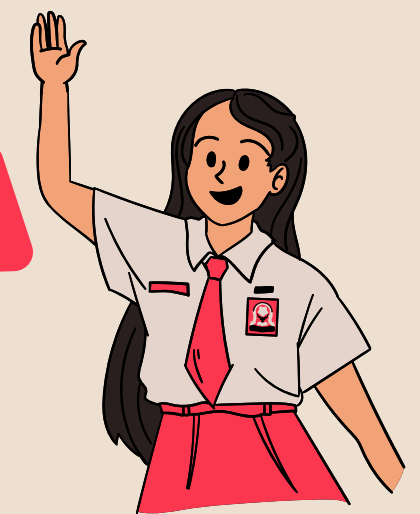
STAKEHOLDERS



CTL

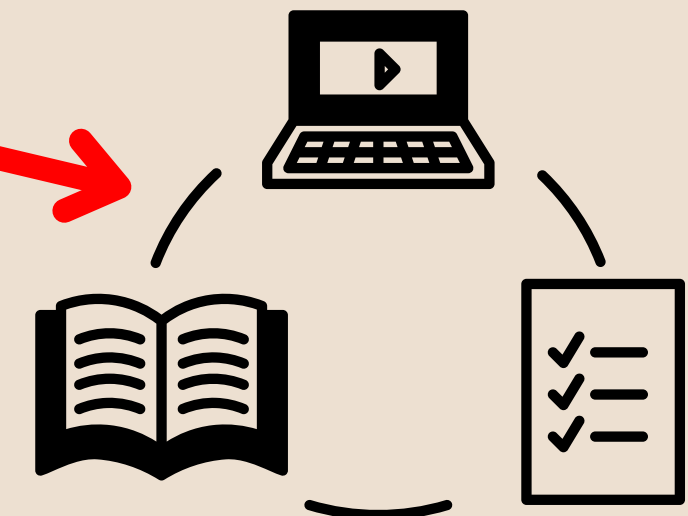


SSE



SIGNIFICANCE

GOAL: IMPROVE LEARNING EXPERIENCE



RESULTS

INTERVENTION STRATEGIES



STATEMENT OF THE PROBLEM

Expectation:

- Use LMS Data
 - Determine engagement attributes
 - Group students according to professors' engagement attributes
 - Test engagement attributes' impact on students' final grades
 - Suggest adjustments to professors' engagement with the LMS to maximize students' final grades

Reality: CTL lacks knowledge on the techniques required for the tasks

RESEARCH OBJECTIVE

Utilize LMS engagement data to **cluster** students, **predict** their final grades, and determine the **optimal combination** of LMS features that will maximize their final grades

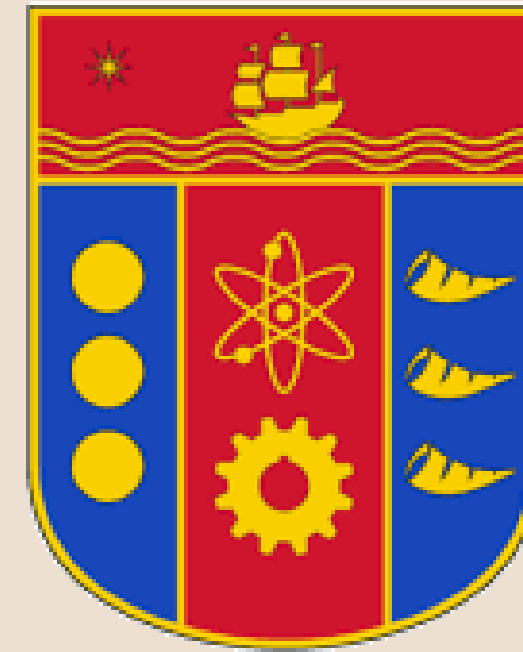
DATASET SOURCE & DESCRIPTION



SCOPE AND LIMITATIONS



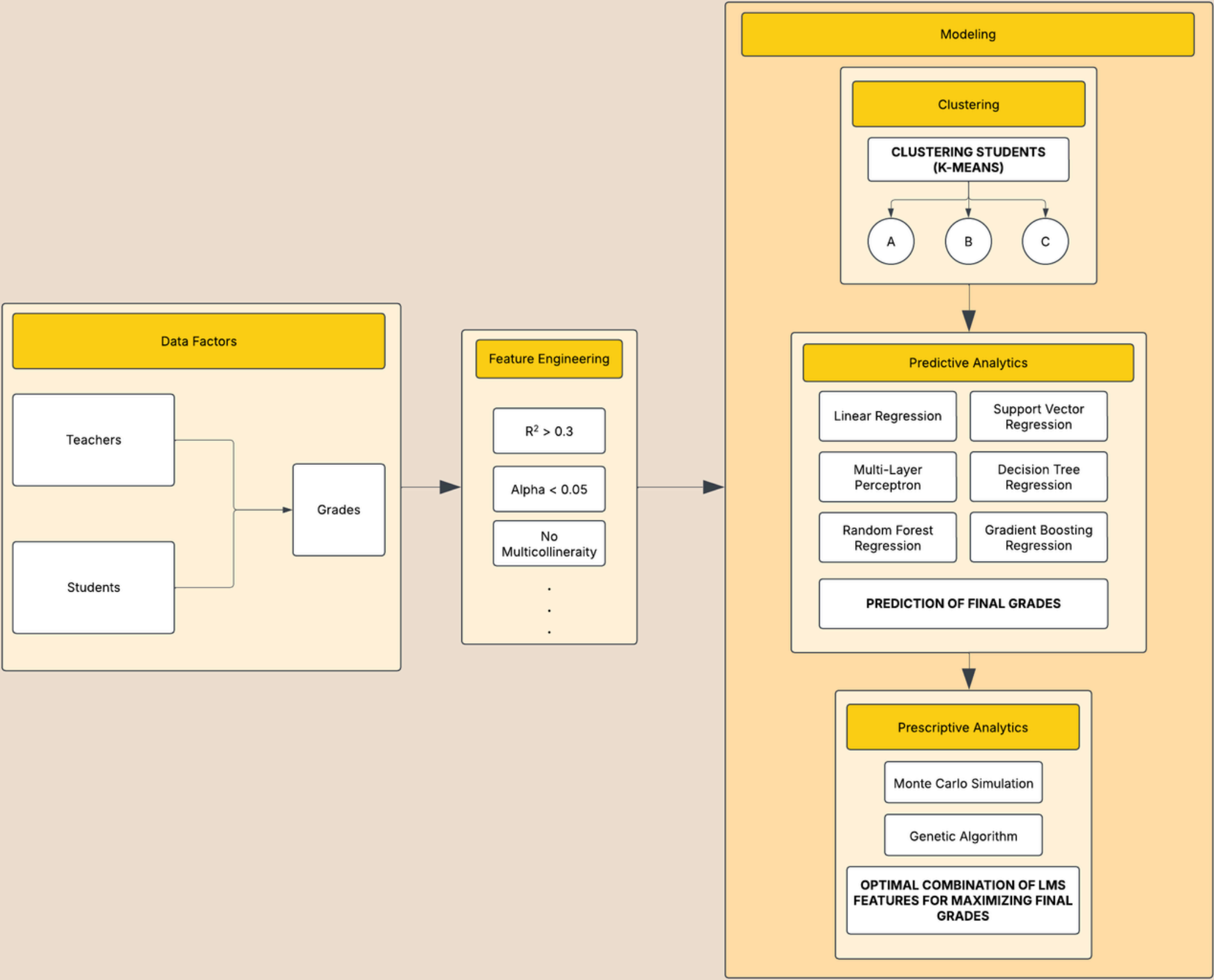
ONLY LMS Data



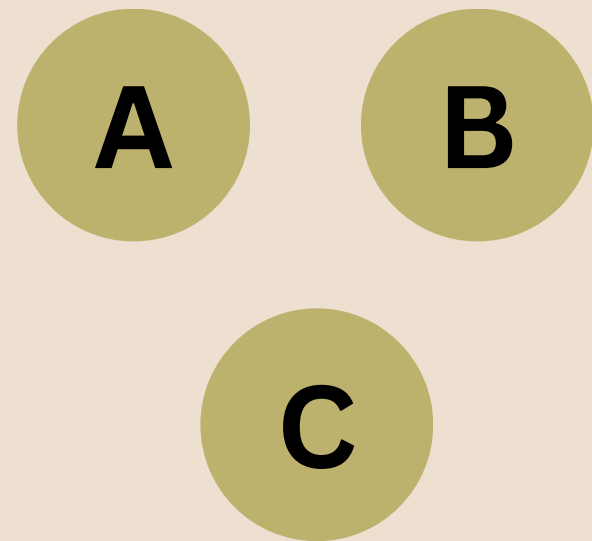
**ONLY SSE
courses data**

A.Y. 2023 - 2024

METHODOLOGY



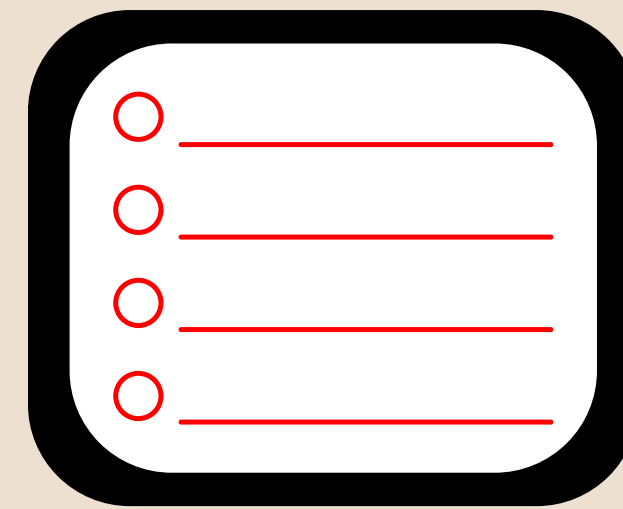
SOLUTION REQUIREMENTS



**STUDENT
CLUSTERS**

$> 30\%$

R² FROM MODELS



**LIST OF
ATTRIBUTES**

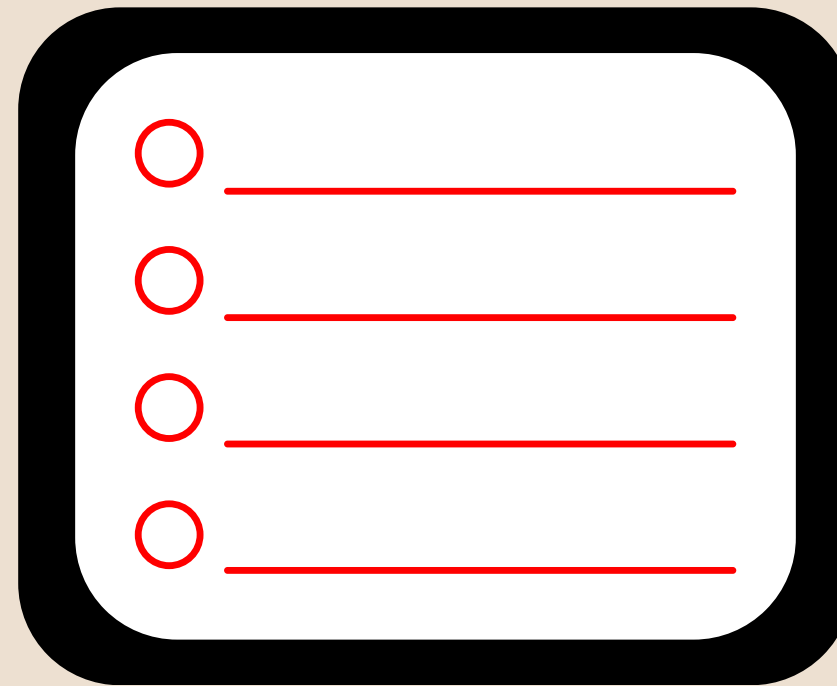


**Optimal Number of
LMS Feature Usage**

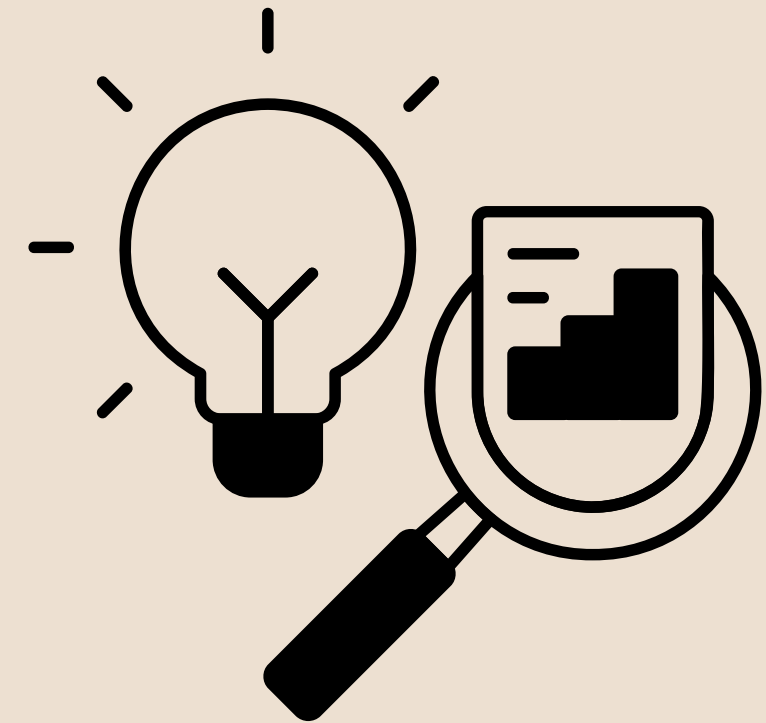
EXPECTED ANALYTICS RESULTS



**MODEL
EVALUATION
RESULTS**



**LIST OF TOP
ATTRIBUTES**



**INSIGHTS
GENERATION**

Review of Related Literature

Title	Identifying Student Behaviour Patterns Based on LMS Data (Levin, 2024)
Objective	Identify student behavior patterns from LMS data through clustering, statistical indicators, and correlation analysis of each cluster with their academic performance
Methodology	1.) Collect data from the Moodle LMS (activity logs, grades and feedback, interaction with content, communications) 2.) Perform K-Means clustering 3.) Calculate statistical indicators for each cluster (average time in LMS (hours), average forum activity (messages), average task completion (percent), average test scores (percent)) 4.) Conduct correlation analysis of each cluster with their average task completion and test scores
Results	1.) Cluster 1 - Hard-working but stressed during tests 2.) Cluster 2 - Balanced high performers 3.) Cluster 3 - Exam masters 4.) Cluster 4 - In need of support
Significance to the Research	1.) Clustering students based on engagement data 2.) Analyzing engagement patterns using statistical indicators 3.) Similarity with predicting final grades based on LMS activity <u>Methodology reference, feature selection, comparison & validation</u>

Title	Data From Flipped Classrooms via Regularization (Yoo, Rho, & Lee, 2022)
Objective	Use Elastic Net (Enet) and Modified Elastic net (Mnet) regularization techniques in the random forest model to predict the final test scores of students in a flipped classroom (FC) setting
Methodology	1.) Collect data (LMS log data, on-grade, on-semester, mean scores from an attitude survey, no. of practicum, final test scores) from 242 pre-service training program students in 8 fully-online classes 2.) Build four random forest models (two with Enet and Mnet regularization, one with tuning on the mtry parameter, one with default settings) 3.) Run the models with five-fold cross-validation 4.) Selection of important variables after regularization
Results	1.) No statistical difference of the four models in terms of prediction 2.) Important predictors: video-watching variables, student attitudes towards the course, number of clicks on software for statistical analysis, number of quizzes taken on PC or mobile
Significance to the Research	1.) Regularization as a technique to be used in Random Forest 2.) Predicting final scores based on LMS data 3.) RMSE as an evaluation metric <u>Reference for random forest, prediction, and prediction evaluation</u>

Title	Effectiveness of LMS Digital Tools Used by the Academics to Foster Students’ Engagement (Simelane-Minsi, 2023)
Objective	Investigate the effectiveness of the LMS digital tools used by academics to foster engagement of students in the LMS
Methodology	1.) Creation of survey and testing for reliability 2.) Testing of survey for reliability 3.) Sending and receiving of survey from two faculties in a university in South Africa 4.) Analysis of answers
Results	1.) Discussion forums, group works, live webinars, Whatsapp and email communication, multimedia learning materials, assessment tools 2.) 90.5% of academics agreed on the effectiveness of the LMS digital tools in fostering students’ engagement; course designs have to be interactive, materials have to be accessible from different devices, collaboration should be encouraged through discussion forums and group works
Significance to the Research	1.) Comparison of LMS features to those found in the study 2.) High engagement in LMS tools correlates with higher learning outcomes 3.) Importance of professor engagement <u>Comparison and validation</u>

Title	Reimagining L&D Capabilities to Drive Continuous Learning (Johnson, 2015)
Objective	1.) <u>Discuss the importance of continuous learning</u> and how the focus of L&D departments hinders it 2.) Capabilities that L&D should focus on to enable continuous learning 3.) Emerging L&D roles 4.) Mindset Shifts
Methodology	
Results	1.) Education - learning while in the classroom (ex. formal lectures) 2.) Experience - hands-on learning (laboratory work, internships) 3.) Exposure - interactions with peers, mentors, industry professionals 4.) <u>Environment - tools, systems, and other infrastructure to support learning (ex. LMS).</u> By Bersin by Deloitte
Significance to the Research	Acknowledgment of limitation of research to just the environment <u>Limitation</u>

Title	The Technological Pedagogical Content Knowledge Framework (Mishra et al., 2014)
Objective	1.) Define TPACK and its components 2.) Go through approaches and issues arising from TPACK 3.) Discuss ways in measuring TPACK
Methodology	1.) Theoretical underpinnings 2.) Relationship between TPACK and educational technology literature 3.) Approaches and Issues from TPACK 4.) Measuring TPACK
Results	Technological Knowledge (TK) - teacher knowledge about traditional and new technologies that can be integrated into curriculum Pedagogical Knowledge (PK) - teacher knowledge about a variety of instructional practices, strategies, and methods to promote students learning Content Knowledge (CK) - any subject-matter knowledge that a teacher is responsible for teaching
Significance to the Research	Professors' pedagogical approaches adapt (PK) to support the integration of the LMS in their teaching (TK), which includes a modification in how the content of the course is delivered (CK) <u>Theoretical support</u>

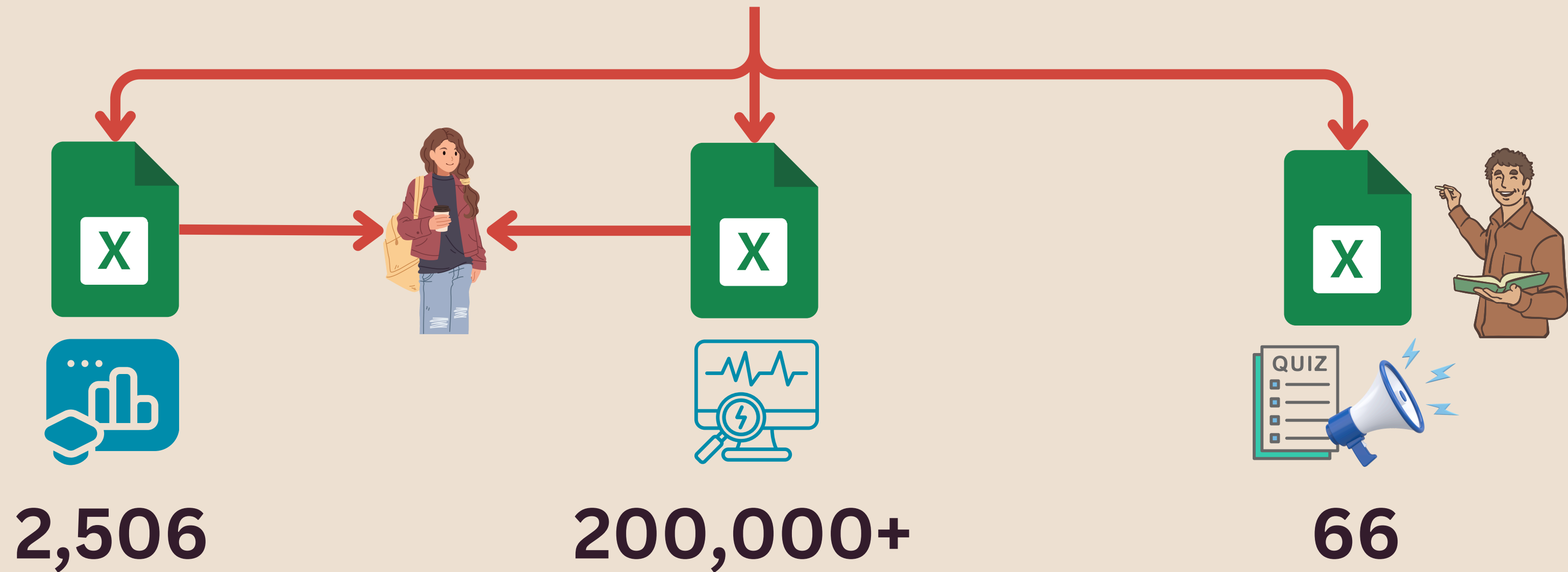
METHODOLOGY

RESEARCH DESIGN

QUANTITATIVE



RESEARCH POPULATION



RESEARCH POPULATION



0

Low
Usage

1

Medium
Usage

2

High
Usage

Announcements

Modules

Assignments with
Submissions

Gradebook

SpeedGrader

Attendance

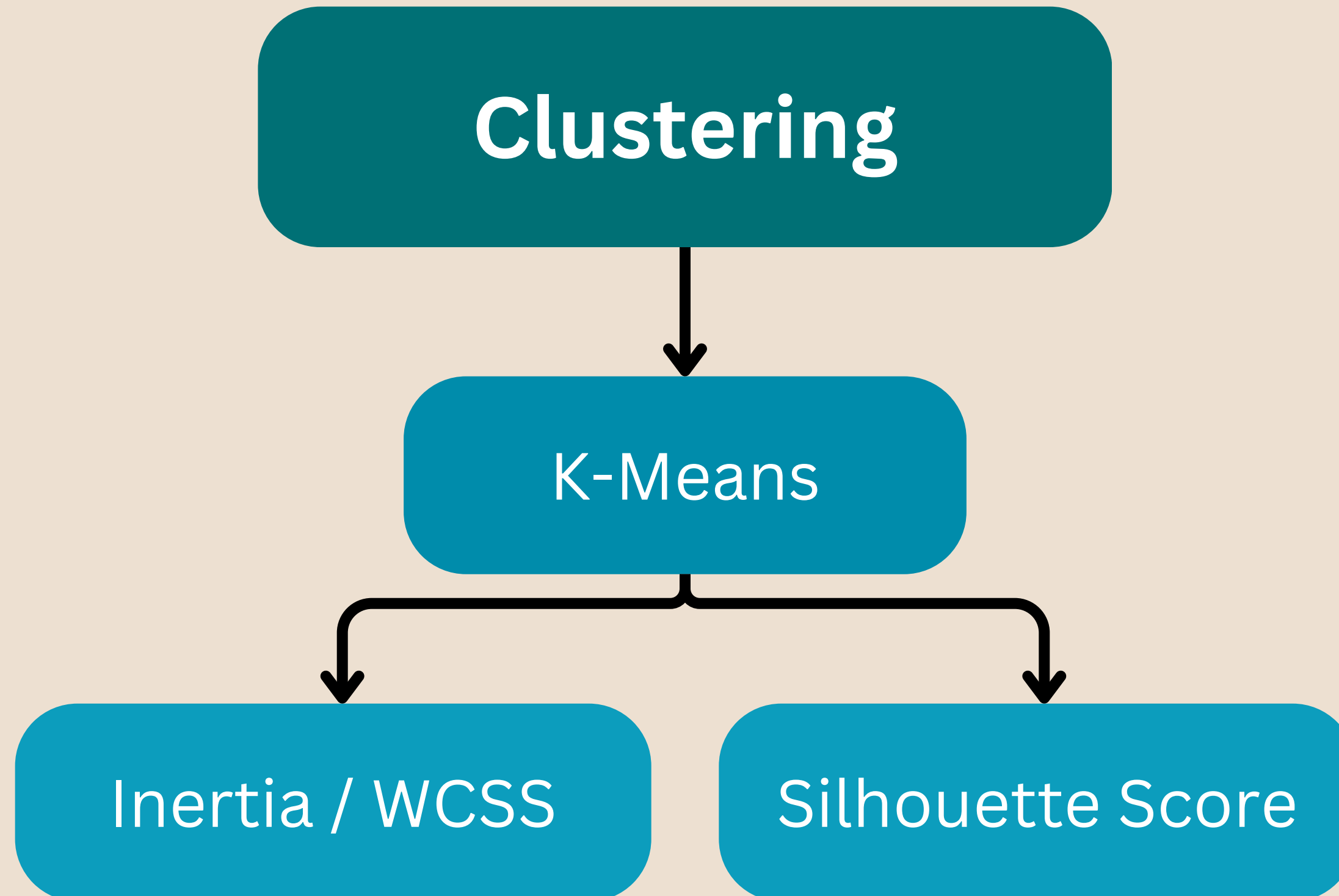
Quizzes

DATA GATHERING

ALL DATA POINTS

- Complete Representation of Academic Performance
- Avoid Information Loss
- Better Training and Generalization
- Consistent Curriculum and Policies
- Controlled Environment

DATA TREATMENT



DATA TREATMENT

Linear Only

**Linear
Regression (LR)**

***With Hyperparameter Tuning and
Cross-Validation**

Linear & Non-Linear

Tree-Based

**Decision Tree Regression
(DTR)
Random Forest Regression
(RFR)
Gradient Boosting Regression
(GBR)**

Others

**Support Vector Regression
(SVR)
Multi-Layer Perceptron
(MLP)**

DATA TREATMENT

Optimization

```
graph TD; A[Optimization] --> B[Monte Carlo Simulation (MCS)]; A --> C[Genetic Algorithm];
```

A flowchart illustrating data treatment methods. At the top is a teal rounded rectangle labeled 'Optimization'. A horizontal line with two downward-pointing arrows branches from the bottom of this box to two separate teal rounded rectangles below. The left box is labeled 'Monte Carlo Simulation (MCS)' and the right box is labeled 'Genetic Algorithm'.

Monte Carlo Simulation (MCS)

Genetic Algorithm

DATA TREATMENT

Linear Regression (LR)

STRENGTHS

- Easier Interpretability
- Performs Well with Linear Relationships

WEAKNESSES

- Assume linear relationship
- Assume normality of residuals
- Outlier sensitivity

DATA TREATMENT

Support Vector Regression (SVR)

STRENGTHS

- Non-Linear Relationships
- Less Sensitive to Outliers
- Can Handle High-Dimensionality

WEAKNESSES

- Unsuitable for Large Datasets
- Less Interpretable
- Lower Performance with More Noise

DATA TREATMENT

Multi-Layer Perceptron (MLP)

STRENGTHS

- Non-Linear Relationships
- Works Well with Large Data

WEAKNESSES

- Prone to Overfitting
- Vanishing/Exploding Gradient
- Computationally Expensive

DATA TREATMENT

Decision Tree Regression (DTR)

STRENGTHS

- No assumption of functional form
- Less Sensitive to Outliers
- Easier Interpretability

WEAKNESSES

- Prone to Overfitting
- High Variance

DATA TREATMENT

Random Forest Regression (RFR)

STRENGTHS

- No assumption of functional form
- Reduces Overfitting
- More robust to outliers

WEAKNESSES

- Harder to interpret than DTR
- Computationally Expensive
- Still Prone to Overfitting without Proper Handling

DATA TREATMENT

Gradient Boosting Regression (GBR)

STRENGTHS

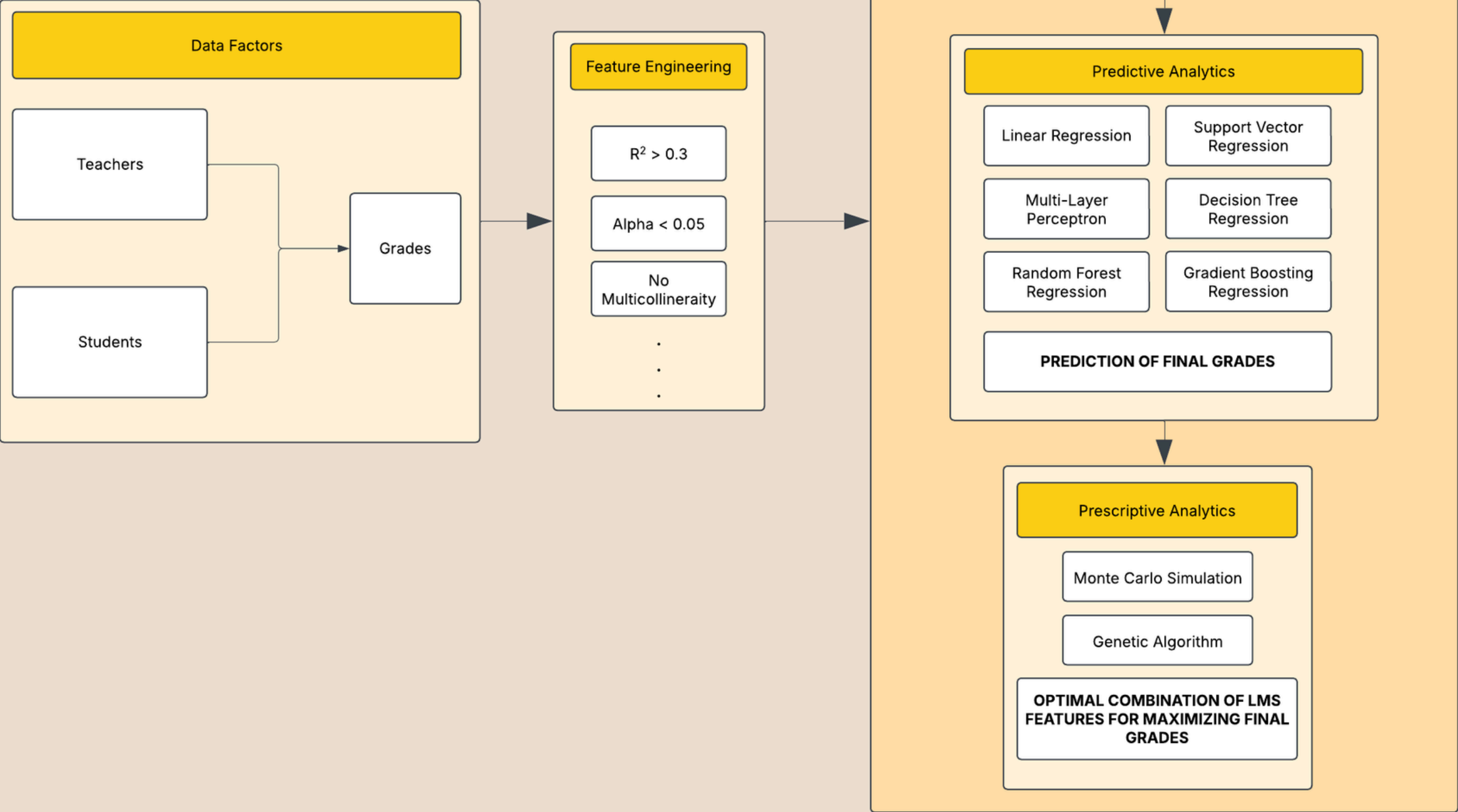
- No assumption of functional form
- Possible higher accuracy
- More robust to outliers than DTR and RFR

WEAKNESSES

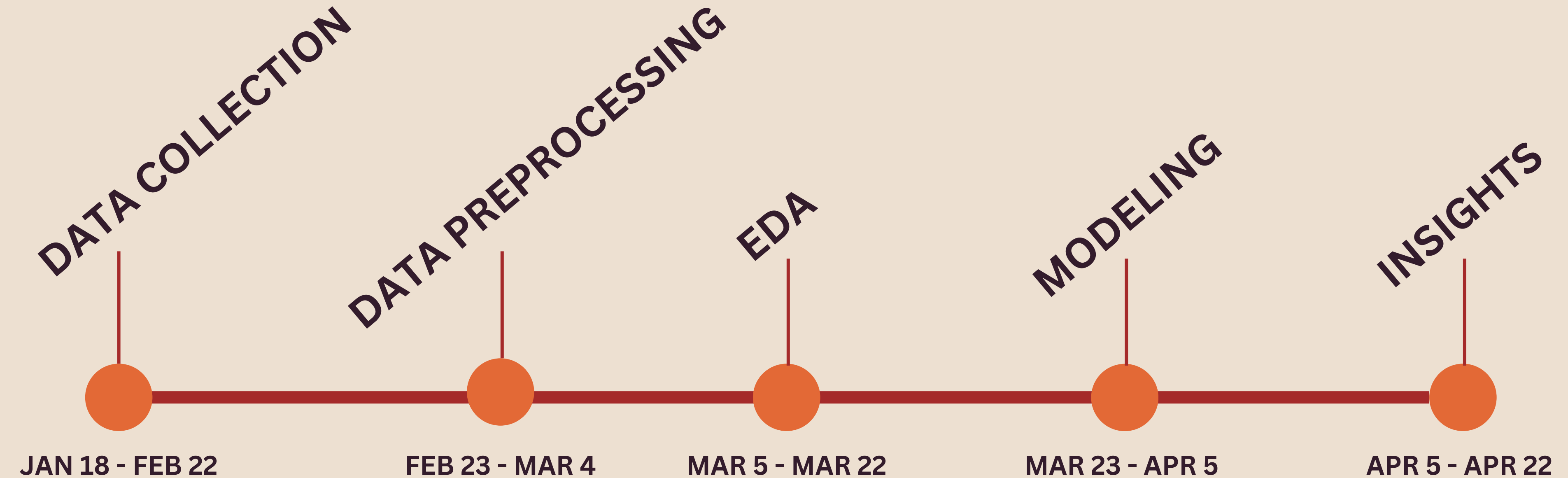
- Harder to interpret than DTR and RFR
- Computationally Expensive
- Higher risk of overfitting without Proper Handling

Metric	Definition	Scale
R2 and Adjusted R2	How much variance in target variable is explained by model; Adjusted R2 <u>guides in adding/removing features</u>	0 - 1
Mean Squared Error (MSE)	<u>Average squared difference</u> between <u>actual values</u> and <u>predicted values</u>	Greater than or equal to 0
Root Mean Squared Error (RMSE)	Square root of RMSE; makes measure in same unit as target variable; <u>easier to interpret</u>	Greater than or equal to 0
Mean Absolute Error (MAE)	<u>Average absolute difference</u> between <u>actual values</u> and <u>predicted values; more robust to outliers</u>	Greater than or equal to 0

EMPIRICAL FRAMEWORK



TIMELINE



THE END